Literature Review

Fairness-Aware Matching and Allocation

DAA Project: Fair and Capacity-Constrained Bipartite Matching

Application: Allocation of college students to projects

Work package status: Initial review and comparison. The final fairness metric is intentionally not selected until the Week 2 team review.

1. Objective

This work package investigates how to allocate students to projects while considering student preferences, project capacities, eligibility constraints, and fairness. The review compares candidate definitions, identifies measurable metrics, and records their computational implications.

The project implementation will progress from maximum-cardinality matching to maximum-weight matching, then capacitated matching using network flow, and finally a fairness-aware formulation selected after the Week 1–2 literature review.

2. Initial bibliography: five credible sources

1. Fair Matching under Constraints: Theory and Applications

Yuichiro Kamada and Fuhito Kojima, The Review of Economic Studies, 2024.

Studies fairness in matching under capacity and other constraints, including justified envy.

https://academic.oup.com/restud/article/91/2/1162/7135693

2. Efficiency and Fairness Criteria in the Assignment of Students to Projects

Nahid Rezaeinia, Julio César Góez and Mario Guajardo, Annals of Operations Research, 2022.

Directly relevant to student-project allocation; examines efficiency and fairness in a bi-objective assignment model.

https://link.springer.com/article/10.1007/s10479-021-04001-7

3. Envy-free Matchings in Bipartite Graphs and Their Applications to Fair Division

Information Sciences, 2022.

Studies a graph-theoretic envy-free matching definition and algorithms for a specific bipartite matching model.

https://doi.org/10.1016/j.ins.2021.11.059

4. Fair Division of Indivisible Goods: A Survey

Georgios Amanatidis et al., IJCAI 2022.

Surveys envy-freeness, maximin-share fairness, and related indivisible-resource allocation concepts.

https://www.ijcai.org/proceedings/2022/756

5. A Review on Fairness in Machine Learning

Dana Pessach and Erez Shmueli, ACM Computing Surveys, 2022.

Provides background on individual and group fairness, measurement, and mitigation; it is not matching-specific.

https://doi.org/10.1145/3494672

3. Common mathematical model

Let S be the set of students and P the set of projects. Let c_j be the capacity of project j, $x_{ij} = 1$ when student i is assigned to project j (and 0 otherwise), and u_{ij} be the utility student i receives from project j.

Each student receives at most one project:

$$\sum_{j \in P} x_{ij} \leq 1 \text{ for every student } i$$

Each project respects its capacity:

$$\sum_{i \in S} x_{ij} \leq c_j \text{ for every project } j$$

For ineligible student-project pairs, set $x_{ij} = 0$. These are feasibility constraints, not fairness metrics.

4. Four fairness approaches

A. Envy-freeness

Definition: A student should not prefer another student's assigned project to their own outcome. In a simplified model, require $u_i(M) \geq u_i(M_k)$ for every pair of students i and k, where M_k is the project assigned to k.

- Protects: Individual students against perceived disadvantage relative to another student's allocation.
- Data: Student preferences/utilities and final assignments.
- Implementation: Count pairwise violations; enforcing the condition may require additional constraints or specialized algorithms.
- Classification: Hard constraint or post-processing metric.
- Applicability: One-to-one and capacitated settings, with a carefully defined envy relation.
- Complexity: Counting violations is straightforward; finding an envy-free allocation may be difficult or infeasible for some instances.

Caution: Preference-based envy is not identical to graph-theoretic envy-free matching definitions used in every paper.

B. Group representation parity

Definition: Students in designated groups should receive comparable assignment outcomes under a specified measure. For group g, define r_g = assigned students in g / total students in g. For two groups, $\Delta = |r_1 - r_2|$.

- Protects: Groups that might systematically receive fewer assignments.
- Data: Group membership, group sizes, and assignment outcomes.
- Implementation: Calculate group rates; optionally minimize the gap or constrain it to a threshold.
- Classification: Post-processing metric, constraint, or objective.
- Applicability: One-to-one and capacitated settings.

- Complexity: Measurement is simple; enforcing parity alongside capacities and eligibility may require ILP or constrained optimization.

Ethical note: Use group attributes only when there is a legitimate, approved reason to assess group-level fairness.

C. Max-min fairness

Definition: Maximize the utility of the worst-off student: $\max_M \min_{i \in S} u_i(M)$. A basic formulation introduces z and maximizes z subject to $u_i(M) \geq z$ for every student, in addition to matching constraints.

- Protects: Students receiving particularly poor outcomes.
- Data: Utility for each student-project pair and a defined utility for remaining unassigned.
- Implementation: Optimize the minimum utility; lexicographic max-min can successively improve the second-worst and later outcomes.
- Classification: Objective.
- Applicability: One-to-one and capacitated settings.
- Complexity: ILP may be appropriate; difficulty depends on utility definitions and additional constraints.

Example: max-min fairness prefers utilities (3, 3, 4) over (1, 5, 5), because the minimum utility is 3 rather than 1, even though the second allocation has higher total utility.

D. Nash social welfare

Definition: Maximize the product of student utilities, $\max_M \prod_{i \in S} u_i(M)$. When utilities are strictly positive, this is equivalent to maximizing $\Sigma_{i \in S} \log u_i(M)$.

- Protects: Balances overall benefit with attention to students receiving low utility.
- Data: Nonnegative, comparable utility scores; zero utilities need special handling.
- Implementation: Integer/nonlinear optimization or a suitable specialized algorithm.
- Classification: Objective, possibly part of a multi-objective model.
- Applicability: One-to-one and capacitated settings.
- Complexity: Potentially high; the logarithmic objective is nonlinear and a linear reformulation is not always straightforward.

Example: Nash social welfare prefers (4, 4), product 16, over (1, 9), product 9, even though (1, 9) has the higher total utility.

5. Comparison table

Criterion	Envy-freeness	Group parity	Max-min	Nash welfare
Fairness unit	Individual	Group	Worst-off student	All students
Main goal	Reduce envy	Comparable assignment rates	Maximize minimum utility	Balance utility
Form	$u_i(M) \geq u_i(M_k)$	Minimize Δ	$\max \min_i u_i(M)$	$\max \prod u_i(M)$
Data	Preferences/utilities	Group labels/outcomes	Utilities	Utilities
Typical category	Constraint or metric	Constraint/objective/ metric	Objective	Objective
One-to-one	Yes	Yes	Yes	Yes
Capacitated	Yes	Yes	Yes	Yes
Post-allocation measurement	Usually feasible	Easy	Easy	Easy if zeros handled
Main limitation	Envy may not be justified	Equal rates ≠ equal quality	May sacrifice total utility	Depends on utility scale/zeros

6. Candidate metrics from project output

Metric	Calculation / purpose
Assignment rate	Assigned students / total students; measures coverage.
Total utility	$\Sigma u_{ij}x_{ij}$; aggregate efficiency.
Average preference rank	Average rank of assigned projects; preference satisfaction.
Worst preference rank	Worst rank among assigned students; worst assigned preference.
Envy violations	Count pairs violating the chosen envy definition.
Group assignment-rate gap	max r_g − min r_g; group disparity.
Minimum utility	$\min_i u_i(M)$; worst outcome.
Nash social welfare	$\prod u_i(M)$, when defined; balanced aggregate welfare.
Capacity violations	Count projects exceeding capacity; validity check.

Specify how unassigned students are handled for preference-rank metrics and define their utility for utility-based metrics. Report feasibility, efficiency, and fairness separately.

7. Implementation implications

Approach	Likely implementation path	Potential advanced method
Envy-freeness	Calculate pairwise violations	Specialized matching algorithm or ILP
Group parity	Calculate group rates	ILP or constrained optimization
Max-min fairness	Optimize minimum-utility variable	ILP or suitable specialized algorithm
Nash social welfare	Optimize product/log-utility objective	Nonlinear/integer optimization or justified approximation

The project execution sequence is: (1) maximum-cardinality matching, (2) maximum-weight matching, (3) capacitated matching using flow/min-cost max-flow, and (4) fairness-aware optimization after the team selects and justifies the formulation. Include NP-hardness, randomized or approximation methods only when justified by the selected model.

8. Work package checklist

- ☐ Collect 4–5 credible papers or authoritative sources.
- ☐ Study at least four distinct fairness approaches.
- ☐ Document definitions, protections, formulas, data needs and implementation implications.
- ☐ Classify each approach as constraint, objective, post-processing metric or multi-objective formulation.
- ☐ Check one-to-one and capacitated applicability.
- ☐ Create the team comparison table.
- ☐ Identify metrics calculable from project output.
- ☐ Flag potential ILP, LP and specialized-algorithm requirements.
- ☐ Keep the final metric undecided until the Week 2 team review.

9. Week 2 team review: open decisions

- Should fairness primarily protect individuals, groups, or the worst-off students?
- Are ranked preferences sufficient, or are numerical utilities needed?
- Should group attributes or project-specific selection priorities be included?

- Is the primary goal maximum assignment count, preference satisfaction, or a balance of efficiency and fairness?
- Which formulation has a defensible definition, measurable outputs, and manageable computational complexity?

Conclusion: This review identifies four candidate fairness approaches, five initial sources, measurable evaluation metrics, and potential computational implications. It does not select a final fairness metric; that decision is reserved for the Week 2 team review.